cea irfm

M2 internship | Six months

UFO detection and severity assessment in WEST

In tokamaks, UFOs (*Unidentified Flying Objects*) are material particles released from surfaces inside the tokamak, including detached deposits and debris, that can enter the plasma. In WEST, these are tungsten-containing materials, which can increase the radiated power and cool the plasma, potentially causing a disruption (that is a rapid termination of the plasma confinement). We can distinguish three different classes of such impacts: transient cooling with recovery (Class 1), strong cooling followed by delayed disruption (Class 2), and radiative collapse followed by rapid disruption (Class 3) [1]. Convolutional neural networks already detect UFOs in infrared images for WEST post-pulse analysis [2], but in this project, severity assessment uses plasma diagnostic signals to estimate the plasma-impact class. This will allow for more versatile and possibly real-time applications in the future, which is of paramount interest for the whole fusion community. Moreover, this exercise is possible thanks to a human-annotated database that has been built for more than 300 shots for a specific experimental campaign of WEST.

Aim

Improve UFO detection and severity assessment using multiple plasma diagnostics and artificial intelligence, then integrate the methods into an automated pipeline that reports results shortly after each discharge.

Proposed work

Initial literature review Review AI methods for time-series event detection and classification, including weakly supervised approaches [3]. Select a small set of methods suitable for UFO detection and plasma-impact classification using multiple diagnostic signals.

1. Benchmark UFO detection: Apply different methods and models to the WEST UFO database to test their performance and compare them with existing detection methods. Identify the best compromise between detection sensitivity, false alarms and detection delay.

2. Classify UFO events and assess severity: Apply and compare methods for classifying UFO events into plasma-impact Classes 1/2/3 using plasma diagnostic signals. Identify signals and temporal patterns associated with each class. Investigate early estimates of the probability of each plasma-impact class from short windows of preceding measurements.

3. Optimise the choice of plasma signals: Improve and test different configurations of signals and plasma parameters to determine which set of plasma signals is most suitable for UFO detection and severity classification, also in view of a future online application.

4. Develop an automated post-pulse analysis prototype: Automatically collect and prepare the plasma signals after each discharge, then apply the models to detect UFOs and estimate their severity. Produce a short report with the results and relevant signal plots. Test the prototype on past discharges and assess how quickly the report can be available after a discharge.

Candidate profile

M2 in machine learning, applied mathematics or physics; Python and statistics skills.

Supervisors

Feda Almuhisen and Samuele Mazzi (CEA-IRFM).

References

[1] J. Gaspar et al., "Thermal and statistical analysis of the high-Z tungsten-based UFOs observed during the first deuterium high fluence campaign of the WEST tokamak," *Nuclear Materials and Energy*, vol. 41, article 101745, 2024.

[2] E. Grelier, J. Bonnail and X. Courtois, "Automated UFO detection via infrared diagnostics in fusion reactors: Application to the WEST tokamak," *Fusion Engineering and Design*, vol. 221, article 115401, 2025.

[3] A. Petralia, P. Boniol, P. Charpentier and T. Palpanas, "Few Labels are all you need: A Weakly Supervised Framework for Appliance Localization in Smart-Meter Series," *Proceedings of the IEEE International Conference on Data Engineering (ICDE)*, pp. 4386–4399, 2025.